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جامعة الملك عبدالعزيز
كلية الحاسبات وتقنية المعلومات

CPCS499 – Project II

Project Report Part I

QISTAS

Intelligent Legal Assistant for Saudi Labor Law Cases

Group_C03

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List of Abbreviations

Abbreviation	Meaning
API	Application Programming Interface
FR / NFR / SR	Functional / Non-Functional / Security Requirement (as numbered in the CS498 report)
GUI	Graphical User Interface
HRSD	Ministry of Human Resources and Social Development
JWT	JSON Web Token
KB	Knowledge Base
LLM	Large Language Model
MRR	Mean Reciprocal Rank
OCR	Optical Character Recognition
PII	Personally Identifiable Information
RAG	Retrieval-Augmented Generation
REST	Representational State Transfer
RTL	Right-to-Left

1. Summary of Previous Work

1.1 Problem Definition

Access to legal assistance in Saudi Arabia is constrained by the cost of professional consultation and the complexity of labor-law procedures. Workers involved in labor disputes often struggle to identify the relevant articles of the Labor Law, to know which entitlements they can claim, to recognize missing information or documents, and to structure a legally sound claim. Labor disputes are among the most frequently litigated matters in Saudi courts, and the problem falls hardest on people who cannot afford a lawyer.

The non-trivial computing challenge lies in Arabic legal language processing. Arabic is morphologically rich and context-dependent, and statutory Arabic encodes obligations through specific modal expressions that generic models can misinterpret. Retrieving the correct provisions from Arabic statutes and grounding a language model's answer in them, without hallucinated legal references, is a genuine research and engineering problem.

1.2 Aims and Scope

CS498 proposed **QISTAS** (Qanuni Intelligent System for Analyzing Saudi Labor Law), an AI-powered web application with five aims: (1) accept Arabic descriptions of labor cases and supporting documents; (2) analyze them with an AI pipeline (NLP, semantic search and retrieval-augmented generation) that extracts key legal elements, identifies missing elements and retrieves relevant provisions and precedents; (3) produce a complete system design with full requirements; (4) evaluate privacy-preserving deployment of Arabic large language models (LLMs) for legal reasoning; and (5) deliver a design ready for implementation in CPCS499.

Following supervisor feedback, the scope was narrowed to Saudi labor law only. Out of scope: criminal, commercial and family law; binding legal advice; and integration with court systems. The target users are members of the public who cannot afford consultation, junior lawyers, and lawyers working outside their specialization.

1.3 Literature Review

The review (Google Scholar, IEEE Xplore, arXiv, ACL Anthology, SSRN; 2024–2026) was organized into three themes:

- **Linguistic complexity of Arabic legal text.** Al-Qahtani (2026) showed that Arabic modal verbs in the Saudi Labor Law encode strict obligations; ALLaM (Bari et al.,

2025) is aligned to the Saudi context; ALARB (Shairah et al., 2025) showed that models adapted to Saudi court data outperform general multilingual models.

- **Retrieval-augmented generation (RAG) for legal accuracy.** ArabLJP (Mohamed et al., 2026) reduced hallucination by grounding Arabic legal answers in retrieved text, and SmartLaw Advisor (Almutairy et al., 2026) reported 98.67% accuracy for Saudi legal consultation with RAG.
- **Multi-agent reasoning and privacy.** LegalMALR (Li et al., 2025) decomposes vague legal queries to find missing elements; DeepSeek-OCR (Wei et al., 2025) reads complex scanned documents; LegalGuardian (Demir et al., 2025) masks personal data before LLM processing.

The identified gap was the absence of a single Arabic assistant for Saudi labor law that combines retrieval of verified provisions, missing-element detection, privacy protection and claim drafting.

1.4 Design Outcomes

CS498 followed a hybrid waterfall methodology with iterative design reviews and produced: 15 functional requirements (FR1–FR15), 7 non-functional requirements (NFR1–NFR7) and 9 security requirements (SR1–SR9), with a traceability matrix; use-case diagrams (general/registered user, lawyer, administrator and the AI engine); a class diagram; two sequence diagrams (case submission and analysis, and login); an entity-relationship diagram; level-0 and level-1 data-flow diagrams; and a system flowchart. The proposed stack was Next.js, FastAPI, PyArabic, ChromaDB, LlamaIndex, a locally deployed LLM (ALLaM or Qwen 3), DeepSeek-OCR and PostgreSQL, with an anonymization layer that replaces personal data with placeholders such as [EMPLOYEE_1] before AI processing.

Supervisor feedback narrowed the scope to labor law, requested detailed non-functional and security requirements, clearer ER cardinalities, a second sequence diagram and a traceability matrix, and advised starting dataset identification immediately for Semester 2. The design was approved at the end of CS498.

2. Methods and Approaches

2.1 Development Methodology

CS498 used a hybrid waterfall, which suited a design-only project with fixed deliverables. For implementation we adopted an **iterative, evaluation-driven** approach: the system was built in short increments, and each increment was measured on test data before the next design decision was taken. The main reason is that an LLM-based system carries uncertainties that cannot be resolved on paper: how well Arabic retrieval works on colloquial questions, how the model behaves on real cases, and what the free API tier allows. For example, the plan to combine keyword and semantic search was dropped after measurement showed the combination *reduced* accuracy (Section 4.6.2).

Every increment was verified at the appropriate level: automated unit and API tests, a retrieval benchmark, hand-checked sample cases, a blind test on unseen cases, and an automated end-to-end browser test.

2.2 Development Phases

Table 1 lists the phases followed in Semester 2 and Figure 1 places them on the semester timeline.

Table 1: Development phases in Semester 2

Phase	Main activities	Output	Verification
P1 Environment and data verification	Development environment (Linux/WSL, Python, Node.js, PostgreSQL + pgvector); collection of official texts; verification of the knowledge-base file against official sources	Working environment; verified data	Every gap traced to an official repeal or merger
P2 Knowledge-base engineering	Database schema with versioning; loader with change detection; structure-based chunking	497 provisions, 535 searchable chunks	Coverage check: all Labor Law articles 1–245 present
P3 Retrieval and evaluation	Gemini embeddings in pgvector; article-number lookup; keyword search; benchmark of 37 questions	Search service	Hit@1, Hit@5, MRR per method
P4 Case-analysis pipeline	Privacy masking; prompt and JSON schema; grounding check; companion-article expansion; entitlement calculator	REST analysis endpoint	Unit tests; 3 hand-checked cases
P5 Blind testing and refinement	12 unseen cases with an answer key; fixes for gender masking, arithmetic, vague cases	Prompt v2–v4; calculator	Re-test of affected cases

Phase	Main activities	Output	Verification
P6 Web interface	Arabic RTL Next.js interface: analysis, search, articles	Prototype GUI	Headless-browser test
P7 Accounts and case management	Registration/login, roles, My Cases, document upload + OCR, claim draft, PDF/Word export, encrypted vault	FR1–FR5, FR10, FR12–FR14	42 automated tests; end-to-end test
P8 Integration and documentation	Real-API end-to-end run, statistics, this report	Report Part I	—

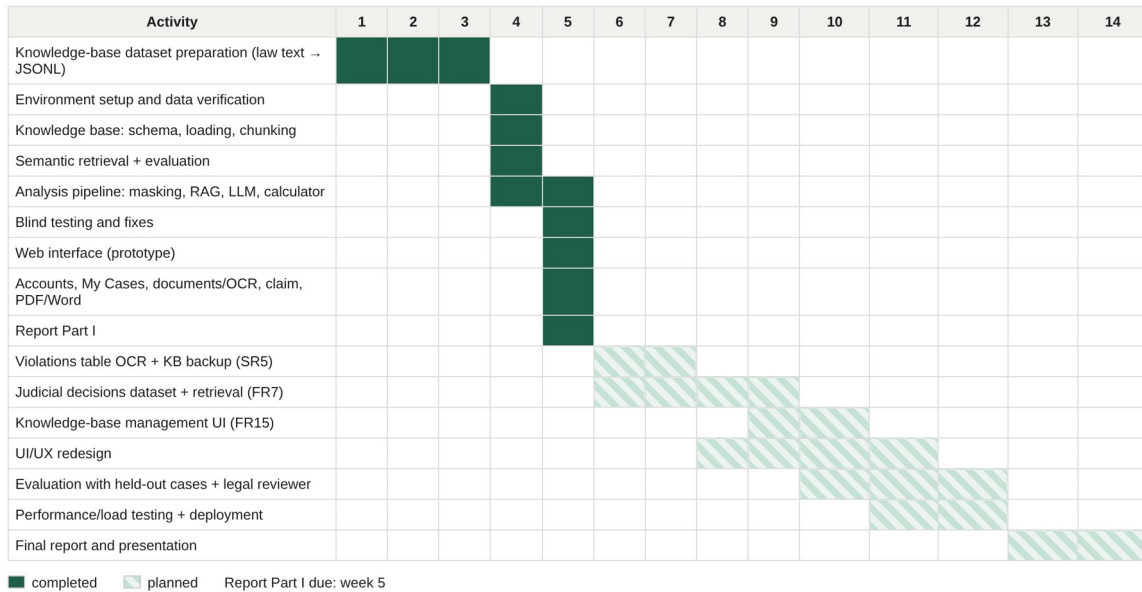


Figure 1: Semester 2 timeline: completed phases and planned work

2.3 Core Technical Approaches

2.3.1 Retrieval-Augmented Generation instead of Fine-Tuning

No labeled dataset of Saudi labor cases (case description → correct analysis) was available, so fine-tuning a model was not feasible. We therefore used RAG (Lewis et al., 2020): for every case the system retrieves the relevant articles from a verified knowledge base and instructs the LLM to answer only from them, citing article identifiers. This needs only the official law text, keeps answers traceable, and lets the knowledge base be updated without retraining.

2.3.2 Knowledge-Base Construction and Verification

The knowledge base contains the Saudi Labor Law, its Implementing Regulations (Ministerial Decision 115921) and five annexes: 456 records extracted from the official publications, one record per article. We checked the file against the official Bureau of Experts text and the HRSD regulations PDF. 38 Labor Law article numbers and 3 regulations articles were absent; each

was traced to an official repeal or merger (for example, Chapter 14, articles 210–228, repealed by Royal Decree M/1 of 22/1/1435H) and stored as a short record stating its status and the responsible decree. The file reflects the 2025 amendments (e.g., 12-week maternity leave, article 151).

Chunking is structure-based: one article is one chunk, because the law is already divided by topic and answers must cite articles. The nine articles longer than 1,500 characters are split at paragraph, line or sentence boundaries, and every chunk carries a header (source, article, chapter) so it is understandable on its own.

2.3.3 Retrieval

Each chunk is embedded with Gemini *gemini-embedding-001* (768 dimensions) and stored in PostgreSQL with pgvector and an HNSW index (Malkov & Yashunin, 2020). A question is embedded the same way and the nearest chunks are returned. Explicit references such as “المادة 77” are resolved directly. Two expansion mechanisms add articles that a correct analysis needs even when the user does not mention them: **companion rules** written from the law's structure (e.g., any termination article brings in the end-of-service award, art. 84) and the **implementing-regulation article** linked to each retrieved Labor Law article. Sources are weighted by legal hierarchy (Law and Regulations above annexes).

2.3.4 Privacy by Design

Before any text leaves the server, personal data is replaced by placeholders: national ID/iqama numbers, phone numbers, IBANs, e-mails, labelled contract or case numbers, names entered in the form (including their short forms) and company names introduced by words such as “شركة” or “مؤسسة”. The database stores masked text only; the real values are encrypted (Fernet) per case and decrypted only for the case owner, implementing SR9. PDFs with a text layer are read locally; scanned documents are sent for OCR only after explicit user consent. The blind test showed that masking a woman's name also hid her gender, which some articles depend on; the form therefore asks for the worker's gender explicitly (Section 3.3).

2.3.5 Grounded Generation and Deterministic Calculation

The LLM receives a strict system prompt (use only the supplied articles, cite them by identifier, state when information is insufficient, never invent articles or amounts) and must answer in a fixed JSON schema: facts, legal issues, cited articles with reasons, analysis, missing information, expected outcome with likelihood, recommended steps and the inputs of each financial entitlement. A grounding check removes any citation that was not retrieved. Money is never calculated by the LLM: it only extracts facts (wage, service dates, contract type), and

code applies the formulas of the Labor Law (arts. 75–77, 84–85, 87, 91, 107, 117, 137), showing each formula to the user. Figure 2 summarizes the pipeline.

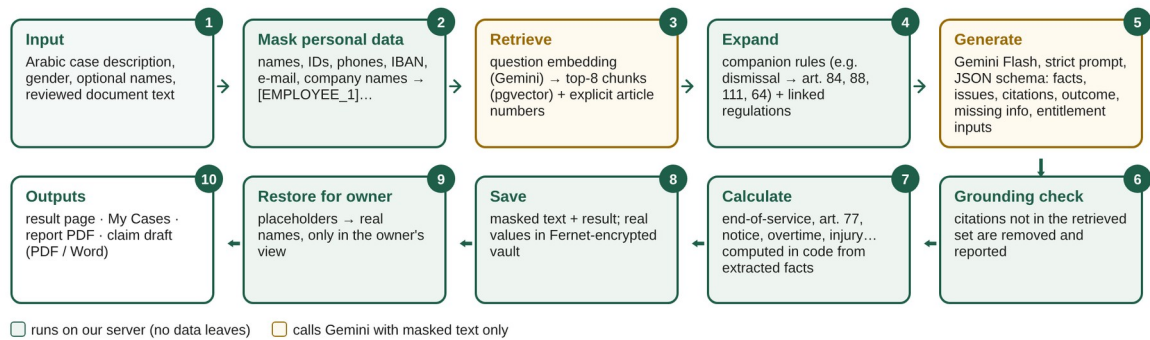


Figure 2: Case analysis pipeline

2.3.6 Verification Strategy

Correctness-critical logic is covered by automated tests that run on a separate test database with the AI service replaced by a fake, so they are repeatable and consume no API quota: masking, the entitlement calculator (hand-checked amounts), grounding, accounts and roles, ownership of cases, document handling, claim drafting and file exports. Retrieval quality was measured with a 37-question benchmark, answer quality with sample and blind cases, and the complete user journey with an automated headless-browser test against the real AI service.

2.4 Changes from the CS498 Design

Implementation led to several justified changes, listed in Table 2.

Table 2: Design changes from CS498 and their justification

CS498 design	Implemented	Reason
Locally deployed LLM (ALLaM or Qwen 3 72B) on a 16 GB RAM + 8 GB GPU workstation	Google Gemini API (Flash family) with masked input, automatic fallback between models	No suitable GPU or budget; a 72B model cannot run on the available hardware. Privacy is preserved by sending only masked text
ChromaDB vector store + relational database	PostgreSQL 18 with the pgvector extension	One database for law, vectors, users and cases; transactions and migrations; one less service to operate
LlamaIndex orchestration	Custom pipeline in Python	Full control over masking, expansion, grounding and calculation; fewer dependencies
DeepSeek-OCR for all uploads	Local PDF text extraction; Gemini OCR for scans with consent	Most digital PDFs need no OCR; consent makes the privacy trade-off explicit

CS498 design	Implemented	Reason
Four roles (user, lawyer, data manager, admin)	Two roles (user, admin)	Prioritized for the 70% milestone; lawyer and data-manager roles are planned
Outputs: analysis report, missing elements, draft claim	Same, plus expected outcome with likelihood and a code-computed entitlement table	Directly answers the users' main question (what am I owed?) with verifiable amounts
Similar judicial decisions retrieval	Not yet implemented	No accessible dataset of Saudi labor rulings yet (Section 3.2)

3. Activities and Actions

3.1 Completed Stages

Table 3 lists the design and development stages that are complete, with the evidence for each.

Table 3: Completed stages and evidence

Stage	Status	Evidence
Requirements and system design (CS498)	Complete	CS498 final report, approved by the supervisors
Data collection and verification	Complete	456 official records; 41 missing article numbers verified as repealed/merged
Knowledge-base database and loader	Complete	497 provisions, 535 chunks, change detection, migrations
Semantic retrieval	Complete	Hit@5 = 100%, Hit@1 = 95% on 37 questions
Privacy layer and encrypted vault	Complete	Masking tests; no plain-text names stored (automated check)
Legal analysis (RAG + LLM)	Complete	Structured analysis with grounded citations; 3 hand-checked cases
Entitlement calculator	Complete	15 calculator tests with hand-checked amounts
Accounts, roles and administration	Complete	Registration, login, 30-minute sessions, admin user management
Case management (My Cases)	Complete	List, view, rename, edit + re-analyze, delete
Document upload and text extraction	Complete	Text PDFs read locally; OCR with consent
Statement of claim draft	Complete	Claim with facts, legal grounds and requests; amounts from code
Downloads	Complete	Analysis report PDF; claim PDF and editable Word file
Prototype web interface	Complete (functional)	Arabic RTL GUI for all features; visual design to be refined
Testing	Partially complete	42 automated tests, blind test, end-to-end test; expert evaluation pending

3.2 Work Not Yet Completed

Table 4 lists the remaining work, the reason it is not complete and the semester week by which we plan to complete it (Report Part I is due in week 5).

Table 4: Remaining work and completion plan

Item	Why not completed	Plan (week)
FR7: similar judicial decisions	No accessible, structured dataset of Saudi labor-court rulings was available during implementation	Collect published labor rulings (Ministry of Justice), index them with the same pipeline (weeks 6–9)
FR15: knowledge-base management UI and data-manager role	Prioritized user-facing features; the knowledge base is currently maintained with a script	Admin screens for articles and sources (weeks 9–10)
2026 Violations and Penalties Table	Official table is a scanned PDF; OCR output must be checked line by line	Extract, verify and load (weeks 6–7)
SR5: daily knowledge-base backup	Not needed for a local prototype	Scheduled database dump (week 6)
NFR1: response time ≤ 10 s for analysis	Free-tier LLM latency (15–90 s, frequent overload)	Progress feedback, caching, evaluate faster models or a paid tier (weeks 10–12)
NFR3 / NFR5: availability and 50 concurrent users	Prototype runs locally; not yet load-tested	Deployment and load testing (weeks 11–12)
SR6: malware scanning of uploads	Basic checks only (file signature, size, page limits, no embedded files)	Integrate an antivirus scanner (week 11)
UI/UX design	Prototype pages are deliberately minimal and replaceable	Full interface design and usability review (weeks 8–11)
Final evaluation	Current test cases were written by the team	Held-out cases written by others and reviewed by a legal practitioner (weeks 10–12)
Lawyer and data-manager roles	Deferred (see Table 2)	Weeks 9–10

3.3 Challenges Encountered and How They Were Resolved

Table 5 summarizes the main problems met during implementation.

Table 5: Challenges and resolutions

Challenge	Resolution
No labeled Saudi labor-case data for fine-tuning	Retrieval-augmented generation over the official law text
Limited hardware (~3.8 GB RAM for the development environment) and no budget	Cloud LLM free tier; open-source PostgreSQL + pgvector; no GPU needed
41 article numbers missing from the knowledge-base file	Each traced to an official repeal/merger and stored as a status record with its decree
Very long provisions ($\approx 11,600$ characters)	Structure-based chunking with boundary-aware splitting and context headers
Keyword search failed on colloquial Arabic (Hit@5 27%) and lowered accuracy when fused	Measured, then adopted semantic search + article lookup (Hit@5 100%)

Challenge	Resolution
Free-tier rate limits (HTTP 429) and model overload (HTTP 503); a model withdrawn (HTTP 404)	Automatic retry with back-off, fallback models, resumable embedding, configurable model names
Retrieval missed entitlements the user did not mention (e.g., end-of-service award)	Companion-article rules and linked implementing regulations
Masking a woman's name hid her gender , which art. 87 depends on (full end-of-service award for a woman resigning within six months of marriage)	The form asks for the worker's gender explicitly; anonymization must preserve legally relevant attributes
LLM arithmetic errors (right rule, wrong amount)	Amounts computed in code from extracted facts, formulas shown to the user
Users give dates instead of durations (“since March 2019”)	The LLM extracts dates; code computes years of service as of the analysis date
Over-confidence on vague cases	Prompt rule: insufficient facts → low likelihood and follow-up questions
Risk of cited articles that were never retrieved (hallucination)	Grounding check removes them; article text is always taken from the database
Short forms of names escaped masking	Name variants masked with the same placeholder
Some PDF generators store Arabic ligatures out of order, so extracted text is garbled	User reviews extracted text and can re-read the file with OCR
The LLM refuses to transcribe published text verbatim (RECITATION)	Detected and reported with a clear message
Evaluation bias: tests written by the same team that tuned the system	Reported transparently; held-out evaluation planned

4. Implementation

4.1 Tools and Technologies

Table 6 lists the tools and technologies used and the benefits that led us to choose them.

Table 6: Tools and technologies with justification

Tool / technology	Used for	Why chosen (benefits)
Python 3.12 + uv	Backend language; dependency and interpreter management	Richest ecosystem for AI and Arabic NLP; uv installs exact, reproducible versions in seconds
FastAPI	REST API	Fast, typed request validation (Pydantic), automatic interactive API documentation (OpenAPI)
PostgreSQL 18	Relational database	Reliable, open source, full Arabic UTF-8 support, transactions; also hosts the vectors
pgvector (HNSW index)	Semantic search	Vector similarity search inside PostgreSQL: no separate vector database to run or synchronize
SQLAlchemy + Alembic	Data access; schema migrations	Database-independent models; versioned, repeatable schema changes for every team member
Google Gemini API (gemini-embedding-001, Gemini Flash family)	Embeddings; analysis, claim drafting, OCR	Strong Arabic quality, structured JSON output, image/PDF understanding, free tier; no GPU required
PyArabic	Arabic text normalization	Removes diacritics/tatweel and unifies letter forms for keyword matching (as planned in CS498)
PyMuPDF	Reading uploaded PDFs; building figures	Fast local text extraction, so text PDFs never leave the server
WeasyPrint	Arabic PDF reports and claims	HTML-to-PDF with correct Arabic shaping and right-to-left layout
python-docx	Editable Word claim	Generates .docx with right-to-left paragraphs that users can edit before filing
cryptography (Fernet), bcrypt, PyJWT	Encrypted vault; password hashing; session tokens	Standard, audited primitives satisfying SR1, SR2 and SR9
Next.js 16, React 19, TypeScript	Web interface	Component-based, type-safe GUI; built-in routing; strong right-to-left support
Tailwind CSS 4 with design tokens	Styling	All visual decisions in one token file, so the interface can be redesigned without touching logic
pytest, Playwright	Automated and end-to-end testing	Repeatable tests with a fake AI service; real-browser testing of the full user journey
Git / GitHub, VS Code + WSL (Fedora Linux)	Version control; development environment	Team collaboration and history; Linux tooling on Windows laptops
Claude Code (AI coding assistant)	Assistance with code, tests and documentation drafts	Faster implementation; all generated code was reviewed and tested by the team (see Appendix A)

4.2 System Architecture

QISTAS follows a three-layer architecture (Figure 3). The presentation layer is an Arabic right-to-left web application. The application layer is a FastAPI service organized in independent modules (authentication, privacy, document reading, retrieval, analysis, calculation, claim drafting and export). The data layer is a single PostgreSQL database holding the legal knowledge base with its vector index, user accounts, masked cases and the encrypted vault. Only masked text is sent to the external Gemini API.

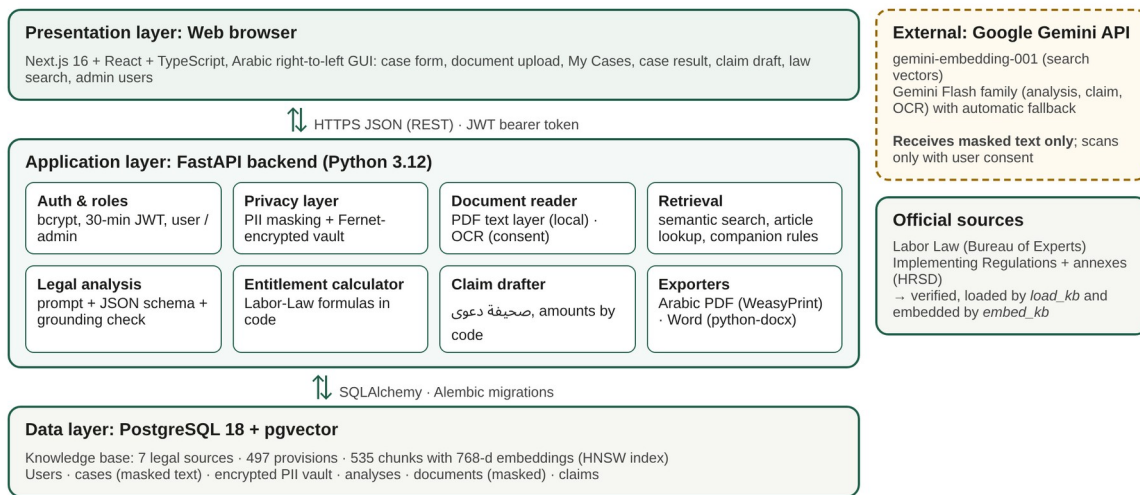


Figure 3: QISTAS system architecture

4.3 Interface Types and Rationale

Table 7: Interfaces developed and why

Interface	Users	Why
Graphical user interface (web, Arabic RTL)	Workers, lawyers, administrators	The target users are non-technical; a web GUI needs no installation, works on phones and computers, and presents long Arabic legal text, tables and downloads clearly
REST API with interactive documentation	Developers; future mobile application	Separates the interface from the logic; every feature is testable from the documentation page (Figure 23) and reusable by a future mobile app (CS498 future work)
Command-line tools	Developers and the data administrator	Loading and embedding the knowledge base, creating administrators, running benchmarks and tests are technical, repeatable tasks best done from scripts (Figure 24)

A menu-driven interface was not developed: the web GUI already provides menu-style navigation (analyze, my cases, search, administration) and suits the target users better.

4.4 Results: Working Interfaces

All screenshots below were taken from the running prototype with a fictitious case (a female accountant dismissed without notice after about 7.5 years of service). Names and numbers are invented.

4.4.1 Accounts

تسجيل الدخول

اسم المستخدم أو البريد الإلكتروني

كلمة المرور

دخول

ليس لديك حساب؟ إنشاء حساب
تنتهي الجلسة تلقائياً بعد 30 دقيقة من عدم النشاط

إنشاء حساب جديد

اسم المستخدم

3-32 حرفاً إنجليزياً أو أرقاماً أو، _ -

البريد الإلكتروني

كلمة المرور

8 أحرف على الأقل

إنشاء الحساب

لديك حساب؟ تسجيل الدخول

(a) Login

(b) Registration

Figure 4: Login and registration pages (FR1, FR2)

4.4.2 Submitting a Case with Documents

قسطاس مساعد قانوني ذي لقضايا نظام العمل السعودي تحليل قضية البحث في النظام قضايا demo_worker تسجيل الخروج

صف حالتك العمالية

وصف الحالة

عملت محاسبة في مؤسسة الأفق للتجارة منذ مارس 2019 بعقد غير محدد المدة، ورثي الأساسي 6000 ريال وبند السكن 1500 ريال في الشهر الماضي أبلغتني الإدارة شفهياً بإنهاء خدماتي بحجة إعادة الهيكلة، دون إشعار مسبق ودون ذكر أي خطأ مني. ما هي حقوقي؟

أذكر ما حدث، ومدة خدمتك، ونوع العقد، والأجر، وما تريد معرفته. كلما زادت التفاصيل كان التحليل أدق.

اسم صاحب العمل أو المنشأة (اختياري)

اسم العامل (اختياري)

جنس العامل

أنثى

بعض الأحكام تختلف بحسب الجنس، ولا يمكن استنتاجه بعد إخفاء الاسم.

مستندات داعمة (اختياري)

PDF أو صورة (JPG / PNG / WEBP)، حتى 10 ميجابايت و10 صفحات، وحتى 3 مستندات

أوافق على إرسال الصور والمستندات الممسوحة ضوئياً إلى خدمة Gemini من Google لقراءتها (ملفات PDF النصية تُقرأ على الخادم ولا تُرسل)

No file chosen Choose File

يُخفى الاسم ورقم الهوية والجوال والأيان تلقائياً قبل إرسال الحالة للذكاء الاصطناعي، ولا تُحفظ البيانات الأصلية.

حُلّ الحالة

Figure 5: Case submission form: description, gender, optional names (FR3)

مستندات داعمة (اختياري)
PDF أو صورة (JPG / PNG / WEBP)، حتى 10 ميجابايت و10 صفحات، وحتى 3 مستندات

☐ أوافق على إرسال الصور والمستندات الممسوحة ضوئياً إلى خدمة Gemini من Google لقراءتها. (ملفات PDF النصية تُقرأ على الخادم ولا تُرسل).

No file chosen Choose File

هذا الملف صورة أو مسح ضوئي ويحتاج موافقتك أعلاه لقراءته.

Figure 6: A scanned image requires consent before it is sent for OCR

مستندات داعمة (اختياري)
PDF أو صورة (JPG / PNG / WEBP)، حتى 10 ميجابايت و10 صفحات، وحتى 3 مستندات

☒ أوافق على إرسال الصور والمستندات الممسوحة ضوئياً إلى خدمة Gemini من Google لقراءتها. (ملفات PDF النصية تُقرأ على الخادم ولا تُرسل).

No file chosen Choose File

إزالة scanned_contract.png قراءة آلية (Gemini) 1 ص

راجع النص المستخرج وصححه إن لزم قبل التحليل:

عقد عمل غير محدد المدة

أبرم هذا العقد بتاريخ 2019/03/01 م بين: مؤسسة الأفق للتجارة (الطرف الأول - صاحب العمل)، والسيدة نورة عبدالله القحطاني، هوية رقم 1076543210 (الطرف الثاني - العاملة).

Figure 7: Scanned contract read by OCR after consent; the text is shown for review (FR4, FR5)

مستندات داعمة (اختياري)
PDF أو صورة (JPG / PNG / WEBP)، حتى 10 ميجابايت و10 صفحات، وحتى 3 مستندات

☒ أوافق على إرسال الصور والمستندات الممسوحة ضوئياً إلى خدمة Gemini من Google لقراءتها. (ملفات PDF النصية تُقرأ على الخادم ولا تُرسل).

No file chosen Choose File

إزالة contract_noura.pdf نص PDF (قُي على الخادم) 1 ص

راجع النص المستخرج وصححه إن لزم قبل التحليل:

عقد عمل غير محدد المدة

أبرم هذا العقد بتاريخ 2019/03/01 م بين: مؤسسة الأفق للتجارة (الطرف الأول - صاحب العمل)، والسيدة نورة عبدالله القحطاني، هوية رقم 1076543210 (الطرف الثاني - العاملة).

البند الأول: تعمل الطرف الثاني بوظيفة محاسبة بأجر أساسي قدره 6000

النص يبدو مشوهاً؟ أعد القراءة بـ Gemini

Figure 8: A text PDF is read locally on the server without OCR

صف حالتك العمالية

وصف الحالة

عملت محاسبة في مؤسسة الأفق للتجارة منذ مارس 2019 بعقد غير محدد المدة، ورتبي الأساسي 6000 ريال وبديل السكن 1500 ريال. في الشهر الماضي أبلغتني الإدارة شفهيًا بإنهاء خدمتي بحجة إعادة الهيكلة، دون إشعار مسبق ودون ذكر أي خطأ مني. ما هي حقوقتي؟

اذكر ما حدث، ومدة خدمتك، ونوع العقد، والأجر، وما تريد معرفته، كلما زادت التفاصيل كان التحليل أدق.

اسم صاحب العمل أو المنشأة (اختياري)

اسم العامل (اختياري)

نورة عبدالله القحطاني

جنس العامل

أنثى

▼

بعض الأحكام تختلف بحسب الجنس، ولا يمكن استنتاجه بعد إخفاء الاسم.

مستندات داعمة (اختياري)

PDF أو صورة (JPG / PNG / WEBP)، حتى 10 ميجابايت و10 صفحات، وحتى 3 مستندات.

☒ أوافق على إرسال الصور والمستندات الممسوحة ضوئياً إلى خدمة Gemini من Google لقراءتها. (ملفات PDF النصية تُقرأ على الخادم ولا تُرسل).

No file chosen Choose File

إزالة

contract_noura.pdf

نص PDF (قُرئ على الخادم) 1 ص

راجع النص المستخرج وصححه إن لزم قبل التحليل:

عقد عمل غير محدد المدة
أبرم هذا العقد بتاريخ 2019/03/01م بين: مؤسسة الأفق للتجارة (الطرف الأول - صاحب العمل)، والسيدة نورة عبدالله القحطاني، هوية رقم 1076543210
(الطرف الثاني - العاملة).
البنء الأول: تعمل الطرف الثاني بوظيفة محاسبة بأجر أساسي قدره 6000

النص يبدو مشوهاً؟ أعد القراءة بـ Gemini

يُخفى الاسم ورقم الهوية والحوال والأبواب تلقائياً قبل إرسال الحالة للنكاه الاصطناعي، ولا تُحفظ البيانات الأصلية.

قد يستغرق التحليل من 15 ثانية إلى دقيقة ونصف.

جاري التحليل...

Figure 9: Progress state while the case is analyzed

20

4.4.3 Analysis Results

قسطاس مساعد قانوني ذكي لقضايا نظام العمل السعودي تحليل قضية البحث في النظام قضاياي demo_worker تسجيل الخروج

→ قضاياي

إنهاء عقد غير محدد المدة دون إشعار تعديل العنوان
#26 - إصدار التحليل 1 أكتوبر 2026
المستندات المرفقة: contract_noura.pdf

تحميل التقرير PDF تعديل الوصف وإعادة التحليل حذف القضية

النتيجة المتوقعة
درجة الترجيح: مرتفع

من المتوقع إلزام صاحب العمل بدفع تعويض عن الإنهاء غير المشروع (المادة 77)، وتعويض عن مهلة الإشعار (المادة 76)، ومكافأة نهاية الخدمة (المادة 84)، بالإضافة إلى تسليم شهادة الخدمة. السبب: نظراً لثبوت إنهاء العلاقة العقدية من طرف صاحب العمل دون توجيه إشعار كتابي مسبق ودون الاستناد لأي من الحالات الحصرية الواردة في المادة 80 التي تجزئ الفسخ بلا تعويض.

المستحقات المالية
محسوبة آلياً وفق مواد النظام، وليست من تقدير الذكاء الاصطناعي.

مكافأة نهاية الخدمة (المادة 84)	٣٧,٥٠٠ ريال
مدة الخدمة من 2019/03/01 إلى 2026/09/01 = 7.5 سنة؛ نصف أجر شهر × 5 سنة + أجر شهر × 2.5 سنة = 37,500 ريال (م84)	
التعويض عن الإنهاء لسبب غير مشروع (المادة 77)	٢٨,١٢٥ ريال
مدة الخدمة من 2019/03/01 إلى 2026/09/01 = 7.5 سنة؛ أجر 15 يوماً × 7.5 سنة = 28,125 ريال (م77)	
التعويض عن عدم مراعاة مهلة الإشعار (المادة 76)	١٥,٠٠٠ ريال

Figure 10: Saved case: title, actions, expected outcome with likelihood, and entitlements (FR6, FR8)

المستحقات المالية
محسوبة آلياً وفق مواد النظام، وليست من تقدير الذكاء الاصطناعي.

مكافأة نهاية الخدمة (المادة 84)	٣٧,٥٠٠ ريال
مدة الخدمة من 2019/03/01 إلى 2026/09/01 = 7.5 سنة؛ نصف أجر شهر × 5 سنة + أجر شهر × 2.5 سنة = 37,500 ريال (م84)	
التعويض عن الإنهاء لسبب غير مشروع (المادة 77)	٢٨,١٢٥ ريال
مدة الخدمة من 2019/03/01 إلى 2026/09/01 = 7.5 سنة؛ أجر 15 يوماً × 7.5 سنة = 28,125 ريال (م77)	
التعويض عن عدم مراعاة مهلة الإشعار (المادة 76)	١٥,٠٠٠ ريال
مهلة الإشعار عند الإنهاء من صاحب العمل 60 يوماً (75) – أجر 60 يوماً = 15,000 ريال (م76)	
الإجمالي	٨٠,٦٢٥ ريال

Figure 11: Entitlements computed in code with the formula of each article

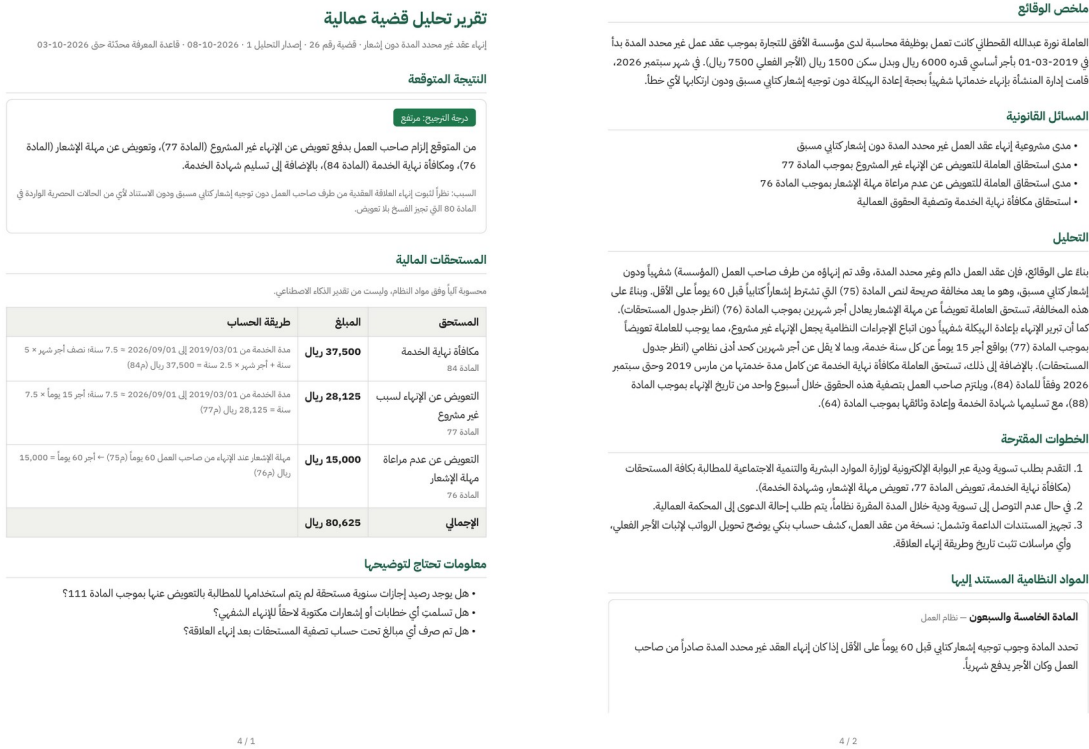
المواد النظامية المستند إليها المادة الخامسة والسبعون نظام العمل تحدد المادة وجوب توجيه إشعار كتابي قبل 60 يوماً على الأقل إذا كان إنهاء العقد غير محدد المدة صادراً من صاحب العمل وكان الأجر يدفع شهرياً. عرض نص المادة
المادة السادسة والسبعون نظام العمل تقرر المادة التزام الطرف الذي لم يرع مهلة الإشعار بدفع تعويض مساوٍ لأجر العامل عن مهلة الإشعار نفسها. عرض نص المادة
المادة السابعة والسبعون نظام العمل تحدد التعويض المستحق للطرف المتضرر من إنهاء العقد لسبب غير مشروع، وهو أجر 15 يوماً عن كل سنة خدمة في العقود غير محددة المدة، على ألا يقل عن أجر شهرين. عرض نص المادة
المادة الرابعة والثمانون نظام العمل تنظم كيفية حساب مكافأة نهاية الخدمة المستحقة للعامل عند انتهاء العلاقة العقدية على أساس الأجر الأخير. عرض نص المادة
المادة الثامنة والثمانون نظام العمل تلتزم صاحب العمل بدفع أجر العامل وتصفية حقوقه خلال أسبوع من تاريخ انتهاء العلاقة العقدية إذا كان الإنهاء من طرفه. عرض نص المادة
المادة الرابعة والستون نظام العمل تلتزم صاحب العمل بإعطاء العامل شهادة خدمة دون مقابل وإعادة جميع وثائقه المودعة عند انتهاء العقد. عرض نص المادة

Figure 12: Cited articles with the reason each applies; the full text expands on demand

ما الذي أرسل للذكاء الاصطناعي؟ بيانات شخصية تم إخفاؤها: 3 عملت محاسبة في مؤسسة [EMPLOYER_1] منذ مارس 2019 بعقد غير محدد المدة، وراتي الأساسي 6000 ريال وبدل السكن 1500 ريال. في الشهر الماضي أبلغتني الإدارة شفهاً بإنهاء خدماتي بحجة إعادة الهيكلة، دون إشعار مسبق ودون ذكر أي خطأ مني. ما هي حقوقتي؟
--

Figure 13: “What was sent to the AI”: the masked text the model received

4.4.4 Downloadable Report



(a) Page 1

(b) Page 2

Figure 14: Downloaded analysis report in Arabic PDF (FR13); the missing-information list and the analysis are on page 2

4.4.5 Statement of Claim Draft

مسودة صحيفة الدعوى

❏نشأ المسودة من التحليل. بيانات الأطراف أدناه لا تُرسل للذكاء الاصطناعي، وتُضاف للمسودة مباشرة.

رقم هوية / إقامة المدعي

جنسية المدعي

جوال المدعي

عنوان المدعي

السجل التجاري / الرقم الموحد للمنشأة

عنوان المنشأة

مدينة المحكمة العمالية

الرياض

إنشاء مسودة صحيفة الدعوى

Figure 15: Claim draft form: party details are inserted by code and never sent to the AI (FR10)

Figure 16: Generated claim draft: parties, facts, legal grounds, requests with computed amounts, notes

مسودة صحيفة دعوى عمالية – أعدت آلياً للمراجعة

فضيلة رئيس المحكمة العمالية بالرياض حفظه الله
السلام عليكم ورحمة الله وبركاته، وبعد:

بيانات المدعي

الاسم	نورة عبدالله القحطاني
رقم الهوية / الإقامة	1076543210
الجنسية	سعودية
رقم الجوال	سعودية
رقم الجوال	
العنوان	الرياض، حي النخيل

بيانات المدعى عليه

اسم المنشأة / صاحب العمل	الأفق للتجارة
السجل التجاري / الرقم الموحد	1010123456
العنوان	

موضوع الدعوى

المطالبة بمستحقات عمالية والتعويض عن إنهاء العلاقة العقدية بسبب غير مشروع

أولاً: الوقائع

1. التحقت المدعية نورة عبدالله القحطاني بالعمل لدى المدعى عليه الأفق للتجارة بوظيفة محاسبة بموجب عقد عمل غير محدد المدة وذلك اعتباراً من تاريخ 01-03-2019م.

2. استمرت المدعية في أداء عملها بانتظام حتى شهر سبتمبر من عام 2026م حين قامت إدارة المدعى عليه بإلغاء خدماتها شفوياً بحجة إعادة الهيكلة.

3. تم الإنهاء دون توجيه أي إشعار كتابي مسبق للمدعية ودون ارتكابها لأي خطأ أو تقصير يبرر هذا الإجراء.

4. امتنع المدعى عليه عن دفع مستحقات المدعية النظامية المتمثلة في مكافأة نهاية الخدمة والتعويضات المقررة نظاماً وتسليمها شهادة الخدمة ووثائقها.

مسودة صحيفة دعوى عمالية – أعدت آلياً للمراجعة

فضيلة رئيس المحكمة العمالية بالرياض حفظه الله
السلام عليكم ورحمة الله وبركاته، وبعد:

بيانات المدعي

الاسم	نورة عبدالله القحطاني
رقم الهوية / الإقامة	1076543210
الجنسية	سعودية
رقم الجوال	سعودية
رقم الجوال	
العنوان	الرياض، حي النخيل

بيانات المدعى عليه

اسم المنشأة / صاحب العمل	الأفق للتجارة
السجل التجاري / الرقم الموحد	1010123456
العنوان	

موضوع الدعوى

المطالبة بمستحقات عمالية والتعويض عن إنهاء العلاقة العقدية بسبب غير مشروع

أولاً: الوقائع

1. التحقت المدعية نورة عبدالله القحطاني بالعمل لدى المدعى عليه الأفق للتجارة بوظيفة محاسبة بموجب عقد عمل غير محدد المدة وذلك اعتباراً من تاريخ 01-03-2019م.

2. استمرت المدعية في أداء عملها بانتظام حتى شهر سبتمبر من عام 2026م حين قامت إدارة المدعى عليه بإلغاء خدماتها شفوياً بحجة إعادة الهيكلة.

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4. امتنع المدعى عليه عن دفع مستحقات المدعية النظامية المتمثلة في مكافأة نهاية الخدمة والتعويضات المقررة نظاماً وتسليمها شهادة الخدمة ووثائقها.

(b) Word (editable)

(a) PDF

Figure 17: Downloaded claim draft as PDF and as an editable Word document (FR13)

4.4.6 Case Management

مساعدة قانوني ذكي لقضايا نظام العمل السعودي

قسطاس

demo_worker تسجيل الخروج

تحليل قضية البحث في النظام قضاياي

تحليل قضية جديدة

قضاياي

إنهاء عقد غير محدد المدة دون إشعار مرتفع مسودة دعوى

آخر تحديث: 18 أكتوبر ٢٠٢٦ إصدار التحليل: 1 إجمالي المستحقات: ٨٠.٦٣٥ ريال

Figure 18: My Cases: the user's saved cases with likelihood, total and claim status (FR12)

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إنهاء عقد غير محدد المدة دون إشعار تعديل العنوان

#26 إصدار التحليل 1 أكتوبر 2026

المستندات المرفقة: contract_noura.pdf

حذف القضية تعديل الوصف وإعادة التحليل تنزيل التقرير PDF

وصف الحالة

عملت محاسبة في مؤسسة الأفق للتجارة منذ مارس 2019 بعقد غير محدد المدة، ورتبي الأساسي 6000 ريال وبديل السكن 1500 ريال. في الشهر الماضي أبلغتني الإدارة شفهيًا بإنهاء خدماتي بحجة إعادة الهيكلة، دون إشعار مسبق ودون ذكر أي خطأ مني. ما هي حقوقتي؟

جنس العامل

أنثى

إعادة التحليل

Figure 19: Editing a case description and re-analyzing it; rename, download and delete actions (FR12)

4.4.7 Law Search and Articles

قسطاس مساعد قانوني ذكي لقضايا نظام العمل السعودي

demo_worker تسجيل الخروج تحليل قضية البحث في النظام قضايي

البحث في نظام العمل ولائحته التنفيذية

إجازة الحج بحث

المادة الرابعة عشرة بعد المائة نظام العمل

للعامل الحق في الحصول على إجازة بأجر لا تقل مدتها عن عشرة أيام ولا تزيد على خمسة عشر يوماً بما فيها إجازة عيد الأضحى ؛ وذلك لأداء فريضة الحج لمرة واحدة طوال مدة خدمته إذا لم يكن قد أداها من قبل. ويشترط لاستحقاق هذه الإجازة أن يكون العامل قد أمضى في العمل لدى صاحب العمل سنتين متتاليتين على الأقل ، ولصاحب العمل أن يحدد عدد العمال الذين يمنحون هذه الإجازة سنوياً وفقاً لمقتضيات العمل .

المادة الثالثة عشرة بعد المائة نظام العمل

مع مراعاة اجازات المرأة العاملة المحددة بموجب هذا النظام، للعامل الحق في إجازة بأجر كامل لمدة (خمس) أيام عند زواجه، أو في حالة وفاة زوجته أو أحد أصوله أو فروع، و (ثلاثة) أيام في حالة وفاة الأخ أو الأخت؛ تحسب جميعها من تاريخ الواقعة، و (ثلاثة) أيام في حالة ولادة مولود له خلال (سبعة) أيام من تاريخ الولادة. وبحق لصاحب العمل أن يطلب الوثائق المؤيدة لهذه الحالات.

المادة العاشرة بعد المائة نظام العمل

1. للعامل بموافقة صاحب العمل أن يؤجل إجازته السنوية أو أياماً منها إلى السنة التالية.
2. لصاحب العمل حق تأجيل إجازة العامل بعد نهاية سنة استحقاقها إذا اقتضت ظروف العمل ذلك لمدة لا تزيد على تسعين يوماً ، فإذا اقتضت ظروف العمل استمرار التأجيل وجب الحصول على موافقة العامل كتابية ، على ألا يتعدى التأجيل نهاية السنة التالية لسنة استحقاق الإجازة .

المادة الخامسة عشرة بعد المائة نظام العمل

1. للعامل - إذا وافق صاحب العمل على انتسابه إلى مؤسسة تعليمية أو قبل استمراره فيها - الحق في إجازة بأجر كامل لتأدية الامتحان عن سنة غير معادة

Figure 20: Searching the Labor Law in everyday Arabic (FR11)



Figure 21: A repealed article shows its status and the repealing decree

4.4.8 Administration

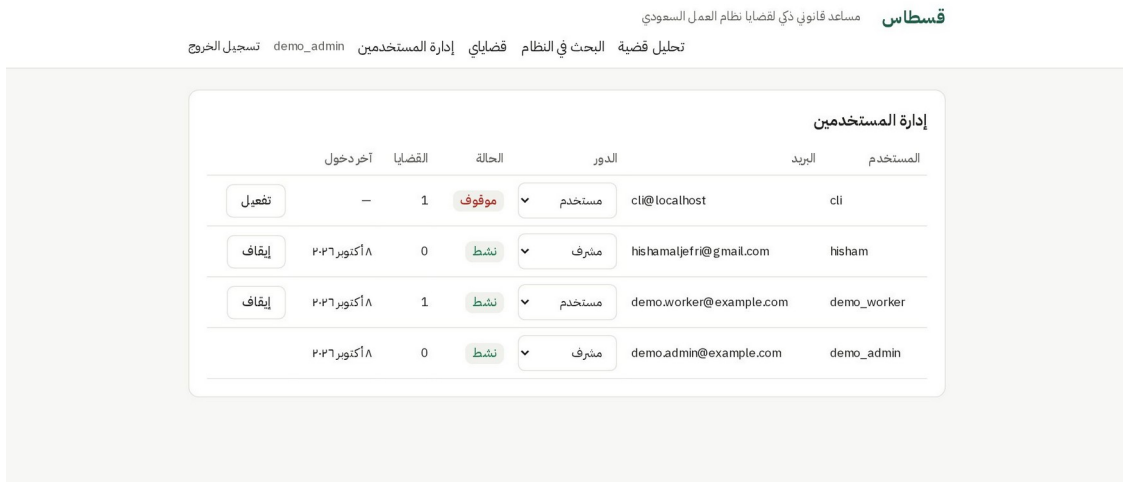


Figure 22: Administrator page: roles and account activation (FR14)

4.4.9 REST API and Command Line

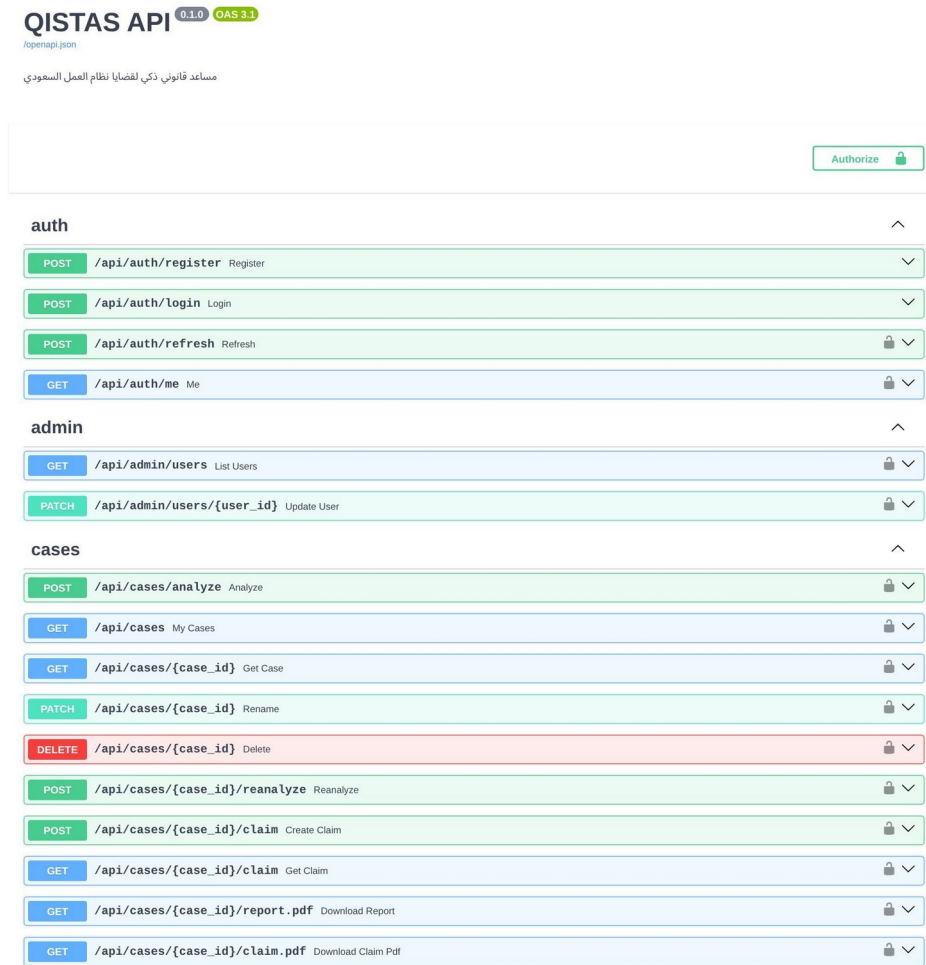


Figure 23: Interactive REST API documentation (excerpt)

```

$ uv run python -m scripts.load_kb
provisions: 0 created, 0 updated, 497 unchanged

$ uv run pytest -q
42 passed in 16.00s

$ uv run python -m scripts.search_cli "كم مدة إجازة الحج؟" -k 3

1. LABOR_LAW | المادة الرابعة عشرة بعد المائة | score=0.01613 | via=vector
للعامل الحق في الحصول على إجازة بأجر لا تقل مدتها عن عشرة أيام ولا تزيد على خمسة عشر يوماً بما فيها إجازة عيد الأضحى ؛ وذلك لأداء فريضة الحج لمرة و

2. LABOR_LAW | المادة التاسعة بعد المائة | score=0.01538 | via=vector
1. يستحق العامل عن كل عام إجازة سنوية لا تقل مدتها عن واحد وعشرين يوماً ، تُزال إلى مدة لا تقل عن ثلاثين يوماً إذا أمضى العامل في خدمة صاحب العمل خ

3. IMPL_REG | المادة (24) | score=0.01449 | via=vector
في تنفيذ أحكام (المادة الثانية عشرة بعد المائة) من النظام: أولاً: تحدد إجازات الأعياد والمناسبات وفقاً لما يلي: 1. إجازة عيد الفطر لمدة أربعة أيام

```

Figure 24: Command-line tools: knowledge-base loader, automated tests and search

4.5 Implementation Completeness

The template requires evidence that at least 70% of the implementation is done. Measured against the CS498 functional requirements, 13 of 15 are complete and one is partial (Figure 25 and Table 8): **87% complete** (90% counting the partial requirement as half).



Figure 25: Status of the CS498 functional requirements

Table 8: Functional requirements status (CS498 FR1–FR15)

ID	Requirement	Status	Evidence
FR1	Register with username, e-mail and password	Complete	Figure 4(b); register endpoint
FR2	Secure login	Complete	bcrypt hashing; 30-minute tokens; tests for wrong, expired and forged tokens
FR3	Authenticated users enter an Arabic case	Complete	Figure 5; analysis requires login
FR4	Upload documents (PDF/image)	Complete	Figures 6–8; PDF, JPG, PNG, WEBP ≤ 10 MB
FR5	Extract text with OCR	Complete	Local text extraction (Figure 8); OCR with consent (Figure 7)
FR6	Extract key legal entities	Complete	Parties/IDs detected by masking (Figure 13); wage, dates, contract type extracted (Figure 11)
FR7	Retrieve similar judicial decisions	Not completed	No dataset yet (Table 4)
FR8	Retrieve relevant labor-law articles	Complete	Figure 12; Hit@5 100% (Section 4.6.2)
FR9	Identify missing legal elements	Complete	Missing-information list (Figure 14)
FR10	Generate a draft statement of claim	Complete	Figures 15–17
FR11	Search the knowledge base	Complete	Figures 20–21
FR12	View, edit and delete own cases	Complete	Figures 18–19
FR13	Download reports and claims	Complete	Figures 14 and 17
FR14	Admin manages accounts and roles	Complete	Figure 22
FR15	Data manager maintains articles	Partial	Loader script with change detection; no UI yet

Tables 9 and 10 report the non-functional and security requirements.

Table 9: Non-functional requirements status

ID	Requirement	Status
NFR1	Response within 10 s	Search: 0.09–0.34 s (meets). Full AI analysis: median 39.8 s (does not meet; see Table 4)
NFR2	Arabic RTL input and output	Met: interface, PDF and Word outputs

ID	Requirement	Status
NFR3	95% availability	Not yet measured (local deployment); automatic model fallback improves AI availability
NFR4	Sensitive data processed locally	Changed by design: only masked text is sent to the LLM; text PDFs are read locally; scans only with consent
NFR5	50 concurrent users	Not yet tested
NFR6	Deployable on 16 GB RAM + 8 GB GPU	Exceeded: runs on a laptop with ≈ 3.8 GB RAM and no GPU
NFR7	Modular design, swappable LLM	Met: model names configurable; layered backend; replaceable interface

Table 10: Security requirements status

ID	Requirement	Status
SR1	Passwords hashed with bcrypt	Met (verified by test)
SR2	Sessions expire after 30 minutes of inactivity	Met: 30-minute tokens refreshed only while the user is active
SR3	No sensitive data sent to external APIs	Changed: masked text only; scans with explicit consent
SR4	Role-based access control	Met: non-admins blocked; users cannot access others' cases (tested)
SR5	Daily knowledge-base backup	Planned
SR6	Uploaded documents checked for malicious content	Partial: signature, size, page and embedded-file checks; files never stored
SR7	Anonymization before AI processing	Met for typed text and text PDFs; scans with consent
SR8	Generalized placeholders such as [EMPLOYEE_1]	Met
SR9	Original values encrypted, owner/admin only	Met: Fernet-encrypted vault per case; no plain names in the database (tested)

4.6 Statistics of Results

4.6.1 Knowledge Base

Table 11: Knowledge-base statistics

Item	Value
Legal sources	7 (Labor Law, Implementing Regulations, 5 annexes)
Provisions	497 (456 official records + 41 repealed/merged status records)
Labor Law coverage	All articles 1–245 plus 4 “bis” (مكرر) articles
Searchable chunks	535, each with a 768-dimension embedding
Knowledge base up to date as of	3 October 2026

4.6.2 Retrieval Accuracy

A benchmark of 37 everyday Arabic questions, each with the correct Labor Law article verified against the official text, was used to compare retrieval methods. Hit@k is the share of questions whose correct article appears in the top k results; MRR is the mean reciprocal rank (Voorhees, 1999). Keyword search performs poorly because users rarely use the statute's wording, and fusing it with semantic results (reciprocal rank fusion; Cormack et al., 2009) lowered accuracy at every weight tested. The adopted configuration reaches Hit@5 = 100% and Hit@1 = 95% (Figure 26). Because the same questions were used to choose the configuration, these figures are optimistic; a held-out evaluation is planned.

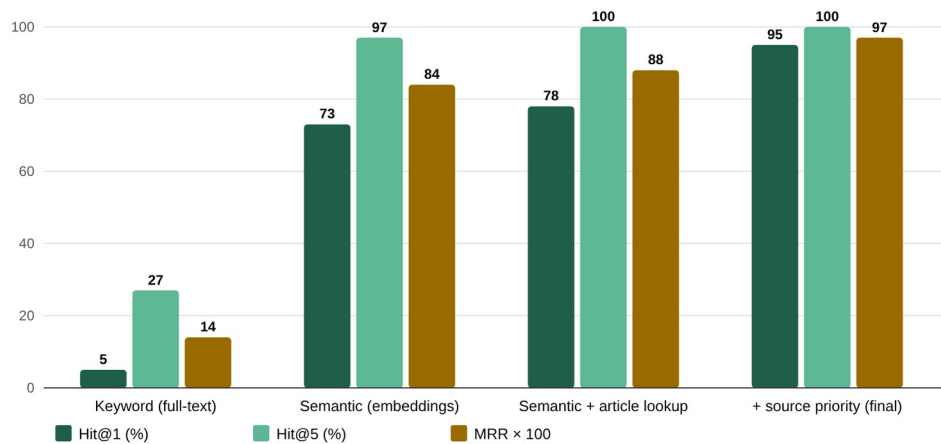


Figure 26: Retrieval accuracy by method (37 questions)

4.6.3 Case Analyses

Table 12: Hand-checked case analyses

Case	Cited articles	Amount computed	Manual check
Dismissal without reason or notice, 6 years, wage 10,000 SAR	75, 76, 77, 84 (+ 80, 88, 64, 111)	85,000 SAR (30,000 art. 77 + 20,000 art. 76 + 35,000 art. 84)	Correct
Wages unpaid for 3 months, 2 years of service	81, 90, 94 (+ 84, 88, 111, 64)	18,000 arrears + 6,000 end-of-service	Correct
Resignation after 4.5 years, wage 7,000 SAR	84, 85, 88	5,250 SAR (one third of 15,750)	Correct
Demonstration case (Section 4.4): dismissal after ≈7.5 years, wage 7,500 SAR	75, 76, 77, 84, 64	37,500 + 28,125 + 15,000 = 80,625 SAR	Correct

A blind test of 12 unseen cases (probation, fixed-term contracts, overtime, leave, sickness, pregnancy, work injury, non-compete clauses, transfers, resignation after marriage, deductions and a vague complaint) gave the following results before the fixes in Section 3.3:

Table 13: Blind test results (12 unseen cases)

Measure	Result
Required articles cited	11 of 12 cases
Conclusion fully correct	8 of 12 cases (4 partially correct)
Invented citations	0
Notable correct behaviour	Told a worker dismissed during probation that nothing is owed (did not side with the user)
Problems found and fixed	Gender hidden by masking; arithmetic error; over-confidence on a vague case; a missing companion article

4.6.4 Automated Testing and Performance

Table 14: Automated tests (all passing)

Test suite	Tests	What is verified
API (accounts, roles, cases, documents, claim, downloads)	16	Authentication, role checks, ownership, encryption, masking end-to-end, uploads, claim grounding, PDF/Word output
Entitlement calculator	15	Every formula against hand-computed amounts, including service years from dates
Privacy masking	9	IDs, phones, IBAN, e-mail, names and short forms, consistent placeholders
Grounding check	2	Invented citations removed and reported
End-to-end browser test (real AI)	12 steps	Register → upload → analyze → My Cases → rename → report → claim → logout → admin → delete; 0 browser errors

Table 15: Performance measurements

Measure	Value
Case analyses recorded	25
Analysis time (median; interquartile range)	39.8 s; 16.1–58.3 s
Analysis time (minimum – maximum)	13.2 s – 88.0 s
Analyses answered by a fallback model	22 of 25 (the primary model was frequently overloaded)
Law search time (5 queries)	0.09–0.34 s (mean 0.17 s)

4.7 User Manual

This manual is written for users with no technical background. The interface is in Arabic; Arabic button names are given in quotation marks.

4.7.1 Creating an Account and Signing In

1. Open the QISTAS website in a browser.
2. Click “إنشاء حساب” (create account), enter a username (English letters or numbers), your e-mail and a password of at least 8 characters, then click “إنشاء الحساب”.
3. Next time, click “تسجيل الدخول” (sign in) and enter your username or e-mail and password.
4. For your security, you are signed out automatically after 30 minutes without activity.

4.7.2 Analyzing Your Case

1. Click “تحليل قضية” (analyze a case).
2. In “وصف الحالة” describe what happened in your own words: your job, when you started, your contract type, your salary and allowances, what the employer did, and what you want to know.
3. Choose the worker's gender (“جنس العامل”); some rights differ for women. Your name and the employer's name are optional and are hidden before analysis.
4. Optional: attach documents such as your contract (PDF or photo). PDF files with text are read on the server. For photos or scanned papers, tick the consent box first. Check the text that appears and correct it if needed.
5. Click “حلّ الحالة” (analyze) and wait; this may take up to a minute and a half.

4.7.3 Reading the Result

- النتيجة المتوقعة (expected outcome): the likely result and how likely it is (high, medium or low).
- المستحقات المالية (entitlements): the amounts you may claim, each calculated by the system with the article and formula shown.
- معلومات تحتاج لتوضيحها (missing information): questions whose answers could change the result. You can add them later with “تعديل الوصف وإعادة التحليل” (edit and re-analyze).

- **المواد النظامية** (legal articles): the articles the analysis relies on; click “عرض نص المادة” to read each one.
- **ما الذي أُرسِل للذكاء الاصطناعي** (what was sent to the AI): your text with personal data hidden.
- The result is for information only and is not legal advice.

4.7.4 Preparing a Statement of Claim

1. On the case page, scroll to “مسودة صحيفة الدعوى” (claim draft).
2. Fill in any details you want in the claim (ID number, nationality, phone, address, the employer's commercial registration number and the court city). These details are added directly and are not sent to the AI.
3. Click “إنشاء مسودة صحيفة الدعوى” and review the draft.
4. Click “تنزيل Word (قابل للتعديل)” to download an editable Word file, or “تنزيل PDF”. Complete any dotted fields, and remember that an amicable settlement request to the Ministry of Human Resources is required before filing at the labor court.

4.7.5 Managing Your Cases

1. Click “قضاياي” (my cases) to see all your saved cases.
2. Open a case to read it again, rename it (“تعديل العنوان”), download the report (“تنزيل PDF التقرير”), edit and re-analyze it, or delete it (“حذف القضية”).

4.7.6 Searching the Law

1. Click “البحث في النظام” (search the law).
2. Type a question or topic in Arabic (for example “إجازة الحج”) or an article number (for example “المادة 84”) and click “بحث”.
3. Click a result to read the full article; repealed articles are marked “ملغاة” with the repealing decree.

4.7.7 For Administrators

1. Sign in with an administrator account and click “إدارة المستخدمين” (user management).
2. Change a user's role or click “إيقاف” to deactivate an account (“تفعيل” re-activates it).

Instructions for installing and running the prototype are given in Appendix B.

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Appendix A: Statement on the Use of AI Tools

An AI coding assistant (Claude Code) was used during implementation to help write and refactor code, tests and documentation drafts, and to automate screenshots. The team defined the requirements and design (CS498), made the design decisions reported here, reviewed the generated code, and verified behaviour through the automated tests, the retrieval benchmark, the hand-checked and blind case evaluations and the end-to-end test described in Section 4.6. Legal content in the knowledge base comes exclusively from the official sources listed in the references.

Note to the team: adapt this statement to the course's academic-integrity policy before submission.

Appendix B: Running the Prototype

The prototype runs on Linux (or Windows with WSL). Requirements: Python 3.12 (managed by uv), Node.js 22, PostgreSQL 18 with pgvector, and a Gemini API key in the project's .env file.

1. Open a Linux terminal in the project folder.
2. Run `./dev.sh`, which starts the database, the backend (port 8000) and the website (port 3000).
3. Open `http://localhost:3000` in a browser; the API documentation is at `http://127.0.0.1:8000/docs`.
4. To make an account an administrator: in the backend folder run `uv run python -m scripts.create_admin --promote USERNAME`.
5. To run the automated tests: in the backend folder run `uv run pytest`.